HEAVY METAL TESTED & CERTIFIED STANDARDS

Heavy Metal Tested & Certified

Program Manual

2026 Edition

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Edition and Notice

2026 Edition

This is the first program-level edition of the Heavy Metal Tested and Certified (HMTc) Program Manual. It describes the program as a whole: why it exists, how it is governed, how a limit is set, how a product is certified against that limit, how the certification is kept honest over time, and how the program is organized to scale across every consumer product category.

It sits above the per-category standards manuals. Each product category that reaches full coverage is published as its own bound volume, the Infant and Child Foods manual being the first in force, carrying that category's master limit tables and per-analyte derivations. This program manual carries the framework those category volumes share. A reader who wants the number for a specific subcategory goes to that category's manual or to the subcategory page on heavymetalcertified.com; a reader who wants to understand the program that produces those numbers reads this volume.

What this manual is, and is not

This manual states how the program works. It is not a marketing document and it does not certify any product. It does not assert that any certified product is safe; it describes a program that tests products against published numeric limits on a defined analyte panel and reports the result. The distinction between an action level and a safety threshold, set out in Part 2.1, runs through the whole document and is load-bearing for its legal and scientific integrity.

The program is deliberately built in two architecturally separate halves, and this manual respects that separation throughout.

The certification program, its limits, and its governance are operated by the Institute of Contaminant Standards and published at heavymetalcertified.com. The peer-reviewed occurrence and toxicology evidence that the limits draw on is compiled openly at the Heavy Metal Index, heavymetalindex.com, which the Institute also maintains. The citation runs one way: the program reads from the Index, and the Index does not endorse the program or any certified product. Part 1.4 describes why that one-way firewall is the source of the program's authority rather than an inconvenience.

Operating documents controlled separately

This manual reproduces the program's foundational operating documents in summarized, reader-facing form. Two of those documents are under independent version control and may be revised between editions of this manual:

- The HMTc Program Governance Policy, the operational constitution, summarized across Part 2.
- The HMTc Lot Testing Schedule, the surveillance and testing protocol, summarized in Part 4.

Where a discrepancy arises between this manual and a controlled operating document, the more recent controlled document governs for operational purposes, and the Program Operator issues a formal Technical Clarification. This edition's summaries are authoritative as of the publication date below.

Methodology continuity

The program rests on a body of published research describing its design rationale. Three peer-track papers, cited throughout and listed in the References, are available on Zenodo: *The Counterproductive Consequences of Public Exposé Testing* (the structural argument for why the program exists); *The Cost of Operating Without Credible Third-Party Heavy-Metal Certification* (the financial argument for the cost of inaction); and *Certification as a Framework for Reducing Heavy Metal Exposure in Infant and Child Foods* (the design rationale for the certification architecture). The structural design principles in those papers remain canonical.

Where a specific calibration value in a paper has since evolved, the current operational value appears in this manual and in the Governing Principles, which supersede the earlier framing for operational purposes.

Publication metadata and rights

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Digital object identifier. A DOI is assigned to this program-level edition on deposit; per-category manuals carry their own DOIs (the Infant and Child Foods manual, for example, resolves to its own registered identifier).

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Intellectual property. The HMTc Certification Mark is pending registration with the United States Patent and Trademark Office (Serial No. 99307306). The heavymetalindex.com corpus that the program's limits draw on is protected under United States copyright and trade-secret law; substantial extraction or re-utilization of the corpus without authorization is prohibited. The pooling engine that computes occurrence-set values is held as a trade secret to the extent described in Part 5.1; its inputs and outputs are auditable under the controlled-terms process there.

How to cite this manual

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For a specific operating rule, append the part and section identifier (for example, “Part 2.11, Standards Ratchet”). For a specific limit value, cite the relevant category manual and its master limit table rather than this program manual, which carries the framework and not the per-subcategory numbers.

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How to Read This Manual

This manual serves four external readers, each opening it for a different reason. The whole volume read end to end gives the complete picture; the reading orders below put the relevant material first.

For brand legal and quality teams

You are evaluating whether the program is defensible enough to recommend, and whether participation increases or decreases your client's litigation exposure. Read Part 1 for the structural and financial argument, Part 2.5 for the Confidential Remediation Track and the two remediation pathways (the chapters most relevant to legal posture, because they convert a finding from a liability into a documented, third-party-verified improvement record), Part 2.7 and Part 2.8 for the analytical and documentary discipline that makes the program's record admissible, and Part 5.2 for the structured response to every category of attack you might face.

For retailer compliance teams

You are deciding whether to recognize the mark as a basis for shelf access. Read Part 2.2 through Part 2.4 (scope, the analyte panel, entry, and the five certification statuses) for what the mark means at each level, Part 4 for the surveillance that backs it, Part 2.10 for the cross-brand supplier-intelligence system that surfaces supply-chain problems early, and Part 5.3 for how the program compares to the certifications you may already accept.

For regulators and public-health officials

You are assessing whether this is legitimate industry self-regulation that complements government maximum levels, or a marketing program in the shape of a standard. Read Part 1.4 for the evidence base and the one-way firewall, Part 2.1 for the framing of limits as action levels rather than safety thresholds, Part 3.1 for how a limit is set (the government-maximum default and the disclosed occurrence calibration beneath it), and Part 2.11 for the data-driven mechanism that commits the program to tighten limits as the market improves.

For adversarial readers

You are looking for the gaps: methodology underspecification, a limit that exceeds a legal ceiling, a ratchet that can be gamed, a conflict of interest. Start at Part 5.2, the program-level adversarial read, which names the attacks the program anticipates and answers each on the merits, so you can see quickly which lines have already been addressed. Then verify the evidence-base claim in Part 1.4, test the ratchet in Part 2.11, and read the chain-of-custody and data-integrity language in Part 4.6 and Part 2.8.

Reading the whole volume

Part 1 makes the case for the program and introduces the evidence base. Part 2 is the operational constitution. Part 3 sets out how limits are derived and where the category coverage stands. Part 4 specifies the testing and surveillance. Part 5 documents the methodology defense, the adversarial read, the recognition landscape, the governance structure, and a glossary.

Part 1. Why the Program Exists

1.1 The problem the program addresses

Heavy metals reach food and consumer products through soil, water, ingredients, processing equipment, and packaging. Lead, cadmium, inorganic arsenic, and methylmercury have no established safe threshold of exposure for the endpoints that matter most in children, and several of the metals accumulate in the body over years. Government maximum levels exist for some products and some metals, but coverage is uneven across categories and jurisdictions, and a limit that a single product meets does not describe the exposure a person accumulates across an entire diet or across the many products they use.

Public attention to this problem has grown sharply. A 2021 report by the United States House Subcommittee on Economic and Consumer Policy found measurable concentrations of toxic heavy metals in commercial baby foods and criticized the absence of finished-product limits for most of them. The Food and Drug Administration responded with its Closer to Zero initiative, a multi-year program to set and lower action levels for lead, arsenic, cadmium, and mercury in foods for babies and young children. A wave of consumer litigation followed. The problem is now a permanent feature of the regulatory, retail, and legal landscape rather than a passing news cycle.

The program exists because the instruments that responded to this problem, useful as each is, leave a gap that neither a brand nor a regulator nor a retailer can close alone. That gap has three parts.

1.2 The structural problem: exposé testing does not drive reduction

The dominant public instrument for surfacing heavy-metal contamination is the exposé test: an advocacy group or publication buys products off the shelf, tests them, and publishes a ranked table naming brands. This does real work in raising awareness, but as a mechanism for driving contamination down it has a structural flaw. It is a snapshot, not a surveillance system. It names the products that happened to test high on the day they were bought, with no follow-up, no verification of correction, and no visibility into whether a low result was typical or a lucky lot. It rewards a brand for being absent from the sample rather than for controlling its supply chain. And because it is adversarial, it gives a brand every incentive to litigate the methodology and nothing to gain by engaging with it.

A brand that reads its own name in an exposé table has no constructive move available. Denying the result attacks the messenger; accepting it concedes liability. The instrument that surfaced the problem offers the brand no path to fix it and prove the fix. The result is an adversarial equilibrium in which contamination is publicized but not systematically reduced.

A certification program inverts this. It publishes the number a product must meet before anyone is tested against it, tests continuously rather than once, verifies correction when a lot fails, and gives a brand a documented, forward-moving record in exchange for participation. It converts the brand from an adversary of the testing into a participant in it. That is the structural difference, and it is the reason the program is built as an ongoing certification rather than a periodic report.

1.3 The financial problem: the cost of operating without credible certification

For a brand, the absence of a credible heavy-metals program is not a neutral state. It is an accumulating, mostly hidden liability. In the current environment a brand without a documented reduction program faces the most exposed possible posture in litigation and regulatory inquiry: the position of having known that heavy-metal contamination is an industry-wide issue and having no third-party-verified evidence of having acted

on it. Retailer questionnaires increasingly ask for exactly the documentation such a brand cannot produce. A single exposé result, a single state enforcement action, or a single plaintiff's filing can trigger reformulation, recall, delisting, and defense costs that dwarf the cost of a surveillance program run continuously in the background.

The financial case for certification is therefore not primarily the cost of the mark. It is the difference between two legal and commercial postures: the brand that can show a timestamped, independent record of testing and improvement, and the brand that cannot. The program is designed so that enrolling, even before the mark is earned, moves a brand from the second posture to the first. Part 2.4 and Part 2.5 describe the entry pathways that make this possible, including the confidential track for brands not yet at certification standards.

1.4 How the program knows what it knows

A certification program that sets numeric limits is only as defensible as the evidence its limits rest on. The most damaging attack a plaintiff's expert or a competing certifier can make is that the standards are self-published: that the certifier set a number, cited its own website as authority, and sold a mark against it. The program is built specifically to make that attack unavailable.

The evidence base is the Heavy Metal Index, a public, curated corpus of the peer-reviewed and regulatory literature on heavy metals in food and consumer products, maintained at heavymetalindex.com. Every occurrence figure and toxicological anchor the program relies on traces to a source page there, and each source page summarizes an underlying study or agency document with its concentrations, sample sizes, methods, and basis, and links to the original where available. An adversarial reader who wants to audit a limit can follow it back to the sources that produced it.

The Index and the certification program are architecturally separate properties, and the citation between them runs one way. The certification program reads from the Index; the Index does not endorse the certification program, does not mention certified brands, and does not exist to justify the program's thresholds. This separation is not incidental. It is what lets the program say, credibly, that its evidence base is an independent reference that the operator happens to maintain, rather than

the operator's own marketing. The same architecture is what allows the program to certify to limits tighter than the literature strictly compels without contaminating the literature record: the gap between what the evidence supports and what the program requires is stated openly, because the two live in different places and answer to different standards. The firewall is a standing operating rule of the program; here it is enough to say that the program's authority comes from the independence of its evidence, and the whole architecture is arranged to protect that independence.

Part 2. The Program

Part 2 is the operational constitution of the Heavy Metal Tested and Certified Program. It states the principles every decision is measured against, the analyte panel, the entry and baseline requirements, the certification statuses and what each permits, the rules for using the mark, the decision rules that turn a laboratory result into a status, and the enforcement, ratchet, and appeals machinery. It summarizes the HMTc Program Governance Policy, which is the controlled document of record; where the two differ, the Governance Policy governs.

2.1 Purpose and core principles

The program sets category-specific numeric limits for priority toxic heavy metals in finished products and certifies products against them under continuous, independent surveillance. It is designed to serve five objectives at once, none subordinate to any other.

1. **Drive contamination down through consumer demand.** The mark creates market preference for cleaner products. Limits are set tight enough to reward genuine reduction, not to ratify current contamination levels.
2. **Safeguard consumers.** Every limit is set at the most protective level that is commercially viable for the clean platform. Infant and child populations receive the highest protection. When in doubt, the consumer is protected.
3. **Safeguard brands that enter the program.** Brands that invest in meeting the standards must receive meaningful differentiation. If the bar is so loose that every brand clears it, the mark is worthless; if it changes unpredictably, brands cannot plan. Standards tighten over time but never loosen, and the program offers a reachable path for both clean and difficult subcategories.

4. **Be legally defensible.** Every limit traces to documented inputs. Every regulatory claim cites a real instrument. No program limit exceeds a government maximum. The methodology is reproducible and its arithmetic is held on the workbench side, separate from the public evidence layer.
5. **Become the de facto standard.** The program scales across every product category with one methodology, and the mark is credible enough to function as a primary purchasing signal. Commercial success follows from scale and credibility, not the reverse.

A decision that serves one principle while undermining another is flagged and resolved rather than optimized silently for whichever principle is most convenient. These five are the test applied to every design choice in this manual.

Three framing rules follow from the principles and recur throughout.

Action levels, not safety thresholds. The program's limits are risk-management action levels anchored to occurrence data and regulatory benchmarks. They are expressly not safety thresholds. Exceeding an action level triggers sourcing, process, and quality-control intervention; it does not by itself mean a product is unsafe or that a consumer has been harmed. This distinction is fundamental to the program's scientific and legal integrity and is reflected in every program communication.

ALARA. The program operates under an As Low As Reasonably Achievable paradigm. For metals with no known safe threshold in children, the program treats continuous, feasible reduction as an enforceable obligation, not aspirational language. A brand that certifies and then plateaus is not meeting the program's requirements. ALARA is operationalized through trend monitoring, corrective action, verification of effectiveness, and the Standards Ratchet.

Concentration-based limits. All limits are expressed as concentrations, parts per billion equivalent to micrograms per kilogram, in the product's native basis. The program does not use per-serving limits, which would allow circumvention through serving-size manipulation. Concentration limits give a consistent, testable, comparable basis across every category.

2.2 Scope and the analyte panel

The program evaluates products against limits for eight priority toxic heavy metals, tested as ten analytes because two of the eight are also measured in the speciated form that carries the toxicity. Eight metals are tested on every certified product, in every subcategory, whatever the category.

The eight priority metals are lead (Pb), cadmium (Cd), arsenic (As), mercury (Hg), nickel (Ni), tin (Sn), aluminum (Al), and chromium (Cr). Arsenic is speciated to inorganic arsenic (iAs) and chromium to hexavalent chromium (Cr(VI)), which brings the certified panel to ten analytes. Methylmercury is a conditional eleventh: speciated by reflex only where a credible pathway exists, and bounded by the total-mercury limit rather than published as a standalone number.

Each metal is assigned to one of two tiers by its toxicological profile, and the tier determines whether any exceedance can ever be tolerated.

Tier 1 (zero tolerance for exceedance): Pb, Cd, iAs, MeHg. These metals are carcinogenic with no established safe threshold, neurotoxic with no established safe threshold, or bioaccumulative with long biological half-lives in children. No transitional exceedance is permitted for a Tier 1 metal at any margin. Any confirmed result above 100 percent of the applicable limit escalates immediately.

Tier 2 (variance-eligible): Ni, Sn, Al, Cr. These metals have established tolerable daily intakes, shorter biological half-lives, or primarily acute rather than chronic toxicity, and are controllable through sourcing and process. A transitional exceedance up to 150 percent of the limit may qualify for transitional certification under the conditions in Part 2.5.

Metal	Form evaluated	Tier	Basis for tier
Lead (Pb)	Total elemental lead	1	No safe threshold; neurodevelopmental harm at lowest observed levels; JECFA withdrew its tolerable intake
Cadmium (Cd)	Total elemental cadmium	1	Bioaccumulative, 10 to 30 year half-life; nephrotoxic;

Metal	Form evaluated	Tier	Basis for tier
			IARC Group 1 carcinogen
Arsenic (iAs)	Inorganic arsenic, speciated	1	Genotoxic carcinogen; no safe threshold; EFSA reference point overlaps current dietary exposure
Mercury (MeHg)	Methylmercury, speciated	1	Neurotoxic; crosses the blood-brain barrier; accumulates in developing neural tissue
Nickel (Ni)	Total elemental nickel	2	Chronic tolerable daily intake established by EFSA; controllable via sourcing and process
Tin (Sn)	Total elemental tin	2	Acute gastrointestinal endpoint only; sentinel for packaging integrity; large margin to effect
Aluminum (Al)	Total elemental aluminum	2	JECFA tolerable weekly intake established; controllable via water treatment and process
Chromium (Cr)	Total; Cr(VI) speciated on trigger	2	Cr(III) is an essential nutrient assumed in food; Cr(VI) is an IARC Group 1 carcinogen by ingestion

Speciation upgrades chromium to Tier 1 treatment when a credible hexavalent pathway exists. When speciation is performed, the speciated result becomes the compliance result for that metal, superseding the total-element value. The Technical Committee may reclassify a metal

between tiers on new evidence, with a defined transition period before enforcement, and may add analytes to the panel.

Why the panel is eight, not four. The common industry panel tests four metals: lead, cadmium, arsenic, and mercury. The program tests those four and adds nickel, tin, aluminum, and chromium because a limit is only as protective as the metals it covers, and the four additional metals reach food and consumer products through real, documented pathways that a four-metal panel is silent on. Testing eight and publishing limits for all of them is a deliberate answer to the gap a four-metal panel leaves.

Aggregate exposure across categories. A single product that meets a government action level does not describe the exposure a person accumulates across a diet. A day's consumption of several infant-food categories, each at its individual FDA action level, can yield a lead intake several times the FDA interim reference level for young children. This is the reason program limits sit below government action levels rather than at them, and the reason the program calibrates across categories rather than one product at a time: aggregate exposure, not single-product compliance, is the exposure that matters for the populations the program most protects.

2.3 Program entry and the three-lot baseline

Certification is continuous, and it begins with a baseline rather than a single passing test. Every SKU entering the program is tested across a minimum of three production lots before any status is assigned. A single passing lot certifies nothing.

The three lots must each represent a distinct production run, are each tested for the full panel with speciation where required, are sampled as composites drawn from across the lot, and are documented under chain of custody. The three results are evaluated collectively so that inherent product variability is distinguished from an anomalous lot. Initial status is assigned from the worst-performing lot among the three, subject to the tier rules and the decision rules in Part 2.7.

Initial status assignment. Following the baseline, the Program Operator assigns status from the results: Status A when all three lots are at or below 100 percent of every limit; Status B when one or more lots fall between 100 and 150 percent and the exceedance is confined to Tier

2 metals; Status C when any lot exceeds 150 percent (up to 200 percent) or any Tier 1 metal exceeds 100 percent; and not eligible for certification when any lot exceeds 200 percent or a result indicates a regulatory limit violation. Percent-of-limit (PoL) is defined as the result divided by the applicable limit, times 100; the maximum PoL across all analytes for a lot (MaxPoL) drives the status.

Expedited baseline via prior-data credit. A brand that already operates a mature, accredited lot-testing program may credit up to two of the three baseline lots with qualifying prior results and complete the baseline with a single new full-panel lot. Prior results qualify only when they were produced by ISO/IEC 17025-accredited laboratories at or below the program's limit of quantitation, tested within the preceding twelve months, supplied as structured machine-readable data with chain of custody, expressed in or reliably convertible to the row's native basis, and credited only for the analytes they actually measured. The completing lot is independently sampled and run across the full panel including every analyte the prior data did not cover. This pathway is not available on the confidential remediation track and reduces no per-analyte data requirement for any analyte lacking qualifying prior data.

2.4 The five certification statuses

The program uses a staged compliance framework so that normal variability does not become a commercial catastrophe while any product bearing the mark remains supported by evidence under a defined surveillance protocol. A status attaches to a product, not to a brand, and it is visible on the certificate. Only Status A and Status B products may display the mark.

Status	Meaning	Mark	Condition
A	Certified (full compliance)	Usable	At or below every limit on all ten analytes, native basis (MaxPoL at or below 100 percent)
B	Certified (transitional)	Usable, disclosed	A Tier 2 metal between 100 and 150 percent, un-

Status	Meaning	Mark	Condition
			der the Part 2.5 conditions, with an end date and corrective action; no Tier 1 exceedance
C	Probation	Suspended	A confirmed Tier 1 exceedance, or MaxPoL between 150 and 200 percent, or transitional conditions unmet; elevated surveillance applies
D	Suspension	Prohibited	MaxPoL above 200 percent, or any hard-stop condition (regulatory violation, data-integrity failure); mark removed, stop-ship
E	Revocation	Prohibited	Certification terminated; concealment of a known exceedance is grounds; license terminated

Status C is graduated by severity. Low probation (MaxPoL 150 to 175 percent) requires every-lot testing with hold-and-release, with the brand's own accredited sampling permitted subject to audit. High probation (175 to 200 percent) adds mandatory independent third-party sampling directed by the Program Operator. To exit probation, the brand must produce a defined run of consecutive passing lots at Status A or B levels (a minimum of five, set by the Program Operator) and complete all corrective-action milestones with verified effectiveness. Both conditions must be met.

Reinstatement from suspension and revocation. Reinstatement after Status D requires a completed corrective action with verified effectiveness, a run of consecutive compliant lots set by the Program

Operator, and restored data integrity: not a fresh baseline. Revocation (Status E) requires full re-application to the program at the Program Operator's discretion.

2.5 Transitional variance and the remediation tracks

Transitional certification (Status B) exists so that a brand with a controllable Tier 2 exceedance can enter and improve under the mark rather than being locked out. It is available on Tier 2 metals only, up to 150 percent of the limit, and never for a Tier 1 metal at any margin. A Status B product carries a mandatory corrective action opened within ten business days of the determination, a full plan within thirty days, a verification-of-effectiveness plan showing measurable reduction over subsequent lots, and a default of twelve months to reach Status A. Any lot that exceeds the 150 percent cap, involves a Tier 1 exceedance, or fails to show improvement escalates the SKU to Status C.

The Confidential Remediation Track. A brand whose baseline does not qualify for Status A or B may still enter the program under a confidential remediation track, provided every tested lot meets the brand's applicable legal obligations at the time of testing. Under the track, the brand's enrollment, test data, and remediation progress are treated as confidential business information; the brand receives full access to the program's remediation support, supplier intelligence, and technical guidance; and the brand may not use the mark on any product until at least one SKU reaches Status A or B. The purpose of the track is legal defensibility: participation, whether or not the mark is yet displayed, creates a timestamped, third-party-verified record of proactive improvement, which is the strongest available answer to the most exposed litigation posture, the posture of having done nothing.

The two remediation pathways. When a lot falls short after certification, the brand chooses how the correction is handled. Both pathways require the same corrective action and the same re-testing; what differs is disclosure. Under the confidential pathway, the default, the finding and its correction stay between the brand and the program while the item is open and on schedule, and register visibility, retailer disclosure, and the effect on the certificate follow the published status rules. Under the public pathway, elective, the brand publishes the finding and its remediation

on its own brand page, producing a documented history of catching and fixing a problem rather than an unexplained gap. Neither pathway is a way out of the corrective action, and neither shields a released lot: a released lot confirmed above its applicable limit on product represented as certified is never protected from a required register status change or from retailer, withdrawal, or recall action. A brand that conceals a known exceedance loses its certificate. The choice exists because a program that punished self-reporting would simply stop being told anything, which would make the mark worth less, not more.

2.6 Mark usage rules

Only SKUs at Status A or Status B may bear the mark. Status C, D, and E products, and products on the confidential remediation track, may not display it under any circumstances, and a brand must have at least one SKU at Status A or B before any use of the mark is permitted.

Prohibited claims. The mark certifies that a product was independently tested and met defined action levels under ongoing surveillance. It does not certify safety. A brand using the mark may not make claims that imply absolute safety, the absence of heavy metals, or zero risk. Claims such as “heavy metal free,” “no toxic metals,” or “100 percent safe from heavy metals” are specifically prohibited. The certified scope is a stated variable: the mark speaks to the product, form, and analyte panel actually tested, and not to a different form of the same product or to analytes outside the panel.

Anti-arbitrage. A brand that removes the mark from a SKU during a remediation period cannot resume use after a single passing lot. Re-entry to mark usage requires three consecutive passing lots at Status A or B levels and closure of any open corrective action. This prevents status arbitrage, in which a brand drops the mark during a bad period and re-enters when results improve by chance. A licensee that withdraws from the program entirely and re-enrolls within twenty-four months is handled under the exit-and-re-enter rule in Part 2.11.

Visibility controls. A brand controls whether its certified SKUs appear in any public register or consumer-facing platform. The Program Operator does not publicize a brand’s participation or certified SKUs

without written consent, except as required by an enforcement action or a regulatory obligation.

2.7 Decision rules under measurement uncertainty

A laboratory result is a value with an uncertainty, and the program's pass and fail decisions incorporate the laboratory's reported expanded measurement uncertainty (MU) using a two-sided rule. A lot passes when the result plus MU is at or below the limit, so that even at the top of the uncertainty range the concentration does not exceed the action level. A lot fails when the result minus MU is above the limit, so that even at the bottom of the range it clearly exceeds. When neither condition holds, the result is borderline: the true value could sit on either side of the limit given the uncertainty.

No product is certified on a borderline result alone. Resolution requires at least one of a confirmatory re-analysis of the same extract, confirmatory testing of a new sample from the same lot under chain of custody, or lot hold-and-release pending resolution. The approach is documented, and the more conservative interpretation governs where ambiguity remains.

Two rules protect the integrity of this process. Repeated ad hoc retesting to obtain a passing result is prohibited: any retesting follows a pre-approved, scientifically justified, documented pathway, and undocumented resampling, lab-shopping, or selective reporting is a data-integrity violation. And when a laboratory does not report MU, the result is compared directly to the limit without adjustment; laboratories that cannot report MU are not preferred, and the Program Operator may require transition to one that does.

2.8 Data governance and integrity

The program's ability to demonstrate aggregate improvement over time, which is both the strongest legal defense for an individual brand and the strongest proof that the program achieves its public-health purpose, depends on structured, queryable data. Two requirements follow.

Structured data. Test results must be supplied in structured, machine-readable form, not as PDF documents, so that trend analysis,

statistical evaluation, and population-level aggregation are possible. The brand authorizes the Program Operator in writing to access its laboratory data.

Data integrity as a hard stop. Falsification, manipulation, selective omission, or unauthorized alteration of test data triggers immediate suspension and likely revocation. Integrity violations include submitting falsified results, failing to report required results, using non-accredited laboratories without authorization, breaking chain of custody without documentation, and undisclosed retesting or lab-shopping. The Program Operator audits data integrity as part of routine oversight and may require random verification testing at designated laboratories.

Laboratory requirements. All testing used for certification decisions is performed by laboratories accredited to ISO/IEC 17025 for the relevant methods and food matrices, using methods fit for purpose. The laboratory's limit of quantitation for each analyte must be at or below the applicable program limit; where it is not, a non-detect result is not acceptable proof of compliance and a more sensitive method is required.

2.9 Corrective and preventive action

When a lot falls short, the brand opens a Corrective and Preventive Action (CAPA): a formal, documented process that identifies the root cause and implements both correction and prevention. Timelines scale with severity.

Status	CAPA initiation	Full plan due	Containment
Status B	10 business days	30 calendar days	As needed
Status C	5 business days	15 business days	Within 48 hours
Status D	2 business days	10 business days	Immediate

Every CAPA documents the issue (SKU, lots, analytes, results, limits, and PoL), the impact and historical trend, immediate containment, a structured root-cause analysis covering ingredients, water, process, equipment, packaging, and formulation, short-term corrective actions, long-term preventive actions, and a verification-of-effectiveness plan specifying how many validation lots will be tested and what constitutes success. A CAPA closes only after verification of effectiveness demon-

strates the product has returned to controlled status and the Program Operator formally approves closure.

2.10 Supplier intelligence

The program maintains an anonymized supplier-intelligence system. When a licensee's testing identifies a supplier or ingredient source with elevated levels, the Program Operator may issue an anonymized alert to other licensees using the same supplier or ingredient category, without identifying the reporting brand. Alerts name the ingredient category, geographic origin where known, and the nature of the concern, for example elevated arsenic in rice flour from a specific region. Licensees receiving an alert are expected to evaluate their own supply chain for the flagged risk. Participation is a condition of the license: brands gain the benefit of collective intelligence, and brands outside the program carry the corresponding informational disadvantage.

2.11 The Standards Ratchet

A published limit tightens over time as the certified market improves; it never loosens unless the underlying science changes to justify relaxation. This directional commitment is what makes the mark a forward-moving signal rather than a static gate, and it is specified precisely so that it cannot be attacked as arbitrary or gamed.

Trigger. The ratchet for a given analyte fires only when both conditions hold across two consecutive annual reviews: the 80th percentile of the combined Status A and Status B trigger pool sits at or below 80 percent of the current limit, and at least 50 percent of the SKUs in the pool have lots at or below 60 percent of the current limit. The first condition tests aggregate market shift; the second tests breadth, so the ratchet does not fire on a single large brand pulling the percentile down. Requiring two consecutive reviews confirms sustained improvement rather than a transient event.

Anti-gaming. The trigger pool combines Status A and Status B results so a brand cannot keep its best data out of the calculation by delaying corrective-action closure. And no single licensee may contribute more than 25 percent of the SKUs in the pool for an analyte; a licensee

above the cap has its excess SKUs excluded from the calculation, so the ratchet reflects industry-wide movement rather than one dominant licensee's progress.

Step size. When the trigger fires, the new limit is the greater of the actual 80th percentile of Status A results and 80 percent of the current limit. The second clause is a step cap: no brand faces more than a 20 percent reduction per ratchet event, regardless of how far the market has moved, and residual tightening carries to the next cycle. This converts the ratchet from a potential shock into a predictable, capped progression.

Transition and grandfathering. A tightening carries a transition of not less than twelve and not more than eighteen months from announcement to enforcement. Status A certifications grandfather through the transition; Status B and C SKUs re-evaluate at their next corrective-action cycle against the more stringent of the old and new limits, so the transition window cannot be gamed; Status D and E products are out regardless.

Forward forecast. At every annual review the Program Operator publishes a forward forecast stating the current trigger-pool percentile, the current limit, and the projected next-cycle limit with expected announcement and enforcement timing. This converts the ratchet from a surprise into a planning parameter a brand can model years ahead. Forecasts are projections, not commitments; the Program Operator publishes an annual comparison of prior forecasts to actual outcomes, and a divergence beyond 25 percent triggers a review of the projection method.

Exit and re-enter. A licensee that voluntarily withdraws and re-enrolls within twenty-four months is evaluated against the standard in effect at re-entry, not at exit, and grandfathering applies only to continuous certification. This closes the pathway in which a brand withdraws before a ratchet event to escape the transition and re-enters after it. A licensee that withdraws for documented commercial reasons and re-enrolls after twenty-four months is treated as a new applicant under the three-lot baseline.

2.12 Regulatory escalation

If any tested lot reveals a result that violates an applicable regulatory limit, as distinct from a program action level, in any jurisdiction where

the product is sold, a hard-stop path triggers regardless of the product's certification status: immediate notification of the licensee, a mandatory product hold or market correction, corrective action initiated within two business days, and full cooperation with regulatory authorities. The program does not shield a licensee from a regulatory obligation. Program action levels are set below regulatory limits precisely to keep this scenario from arising, but when it arises, regulatory compliance takes absolute precedence.

2.13 Appeals, reinstatement, and document control

The appeals process is deliberately two-tier to address the institutional conflict between the Program Operator's certification revenue and its role as first reviewer of an appeal.

Tier 1, internal. A licensee may appeal any status decision in writing within ten business days, stating the technical or factual basis and supporting evidence. The Program Operator reviews within fifteen business days and may convene the Technical Committee. The original status remains in effect during review unless temporary relief is granted.

Tier 2, independent. A licensee dissatisfied with the Tier 1 outcome may escalate within ten business days to an Appeals Committee structurally independent of the Program Operator. The committee comprises three external reviewers, none employed by or under contract to the operator: a toxicologist or food-safety scientist, a food-industry quality professional with no financial relationship to the licensee or operator, and a legal adviser with regulatory-affairs background. Members serve staggered two-year terms, sit on no more than three appeals a year, and recuse from any appeal involving a commercial relationship. The committee may affirm, reverse, or remand; a reversal is binding on the Program Operator with retroactive effect to the original determination date. The committee decides within thirty business days, with one fifteen-day extension permitted, and its funding is arranged so that no individual appeal carries financial pressure for the reviewers. The Program Operator publishes anonymized appeal outcomes in the annual transparency report. This independent tier is the structural answer to the conflict an internal-only process cannot fully mitigate.

Document control. Where documents within the program disagree, the Program Operator issues a formal Technical Clarification, and until it does, the more conservative interpretation governs. Individuals involved in certification determinations are free from conflicts of interest with the licensee under review, and the operator maintains impartiality throughout.

Part 3. The Standards

Part 2 described how a product is certified. Part 3 describes what it is certified against: how a numeric limit is set, how the program's categories are organized, and where category coverage currently stands. The per-subcategory numbers themselves live in the category manuals and on the subcategory pages at heavymetalcertified.com; this part carries the method that produces them and the map of what is covered.

3.1 How a limit is set

The program sets a limit for a specific (subcategory, analyte) cell, in the product's native basis. The method has a default and a disclosed exception, and it is the same for every category.

The default is the strictest applicable government maximum. For any analyte, the standard defaults to the strictest maximum level set by a credible government regulator, the European Union, Codex Alimentarius, the FDA, WHO/JECFA, FSANZ, or Health Canada, converted into the row's native basis. This is a lookup, not a computation. It carries the Tier 2 metals directly, and where the law itself splits a limit by formulation, the standard adopts the law's own split and cites it. Defaulting to a sovereign limit is the most defensible possible posture: the standard cites a government's own number, so there is no novel methodology for an adversary to attack, and it is what lets the program scale across every category quickly.

The occurrence calibration runs only where the program sets a number itself. Two kinds of cell have no government maximum to look up: the four Tier 1 toxics, where the program tests whether clean producers can demonstrably beat the government floor, and the cells no government regulates at all. For those cells only, the standard is set from a pooled distribution of peer-reviewed and government occurrence data.

For a Tier 1 toxic the pooled value is floored against the government maximum, so the published number is the stricter of the two and the ratchet is preserved; for an unregulated cell the pooled value stands alone because a lookup returns nothing.

Within the occurrence-set cells, subcategories are paired where one form structurally carries a higher contamination burden than its partner, rice cereal against non-rice, soy formula against cow-milk, root vegetables against above-ground. The pair is kept separate rather than pooled, because pooling would raise the clean limit and flatter the dirty one in the same move. The clean form of a pair, and any subcategory with no within-pair partner, is set at the 97th percentile of its pooled distribution; the structurally higher-burden form is set at a far tighter 45th percentile, because the purpose there is to separate the better operators inside a difficult category rather than to bless the category.

The percentile is a disclosed calibration choice. A brand certifies a row of ten analytes at once, so a per-analyte limit set at a raw 90th percentile would produce a joint row pass rate near 35 percent under independence. The 97th and 45th percentiles are the values that deliver the program's stated row-level outcomes for occurrence-set cells: roughly 90 percent of clean-subcategory brands and 40 percent of dirty-subcategory brands pass on day one. Those targets were confirmed at 94 percent and 42 percent against FDA per-sample lot data. The program publishes this pass rate rather than concealing it, because every percentile-based food-safety limit embeds a pass rate, Codex maximum levels, EU regulation, and FDA action levels among them, and the only choice is whether the body setting the limit tells you what it is.

The health-based floor binds first. No published value ever exceeds the lowest applicable regulatory ceiling in the row's basis. The calibration operates only in the space beneath that ceiling, where the literature alone does not fix a single number. A subcategory whose every published limit binds on a government maximum has no percentile-set cell at all; infant formula powder is one such subcategory, with all eight of its published values set by government lookup.

Publication gates. A per-analyte value publishes only when it clears every gate: a sufficient occurrence pool for the cells set from occurrence, with no single survey exceeding half the pool; every contributing source converted into the row's native basis with the conversion recorded; the

strictest applicable government maximum loaded and expressed in the row's basis; analytical feasibility, so the value is measurable given the limit of quantitation; and exactly one rationale tag. A cell that fails any gate stays unpublished rather than shipping with a caveat.

Rationale tags. Every published value carries one of four tags, stated honestly. `regulatory-alignment` when a government ceiling binds. `literature-baseline` when a pooled occurrence percentile binds beneath the ceiling. `feasibility-driven` when the limit of quantitation binds above the pooled value, paired with a plan to tighten as methods improve. `precautionary` when the program deliberately sets a value below both the pooled percentile and the ceiling, for a documented vulnerable-population or cumulative-exposure reason. Where an occurrence pool exists beneath a binding ceiling, the pooled value is shown alongside the published value so the gap between what the literature supports and what the program requires is visible rather than hidden.

Native basis. Every standard is published in the form the product is actually sold in: powder as placed on market, ready-to-feed as consumed, dry cereal as sold, purée as consumed. Sources reporting in another basis are converted inward during pooling using documented factors, never the reverse. A cross-basis equivalent shown next to a value is a reading convenience, not a second standard a brand may choose between.

3.2 The category taxonomy

The program is organized by a comprehensive testing taxonomy whose organizing principle is exposure pathway crossed with regulatory jurisdiction, with user population (ages zero to five versus general) as a secondary routing variable only where regulation itself distinguishes it. The taxonomy defines an envelope of thirty-five category numbers, of which thirty-four are active; category 7 has been retired in place, its members re-homed to the specialty-food categories 30 through 35. For browsing and citation the categories group into navigational domains, which change no category number or identity.

Foods (ingestion). Category 1 Infant and Child Foods; 3 Grains, Cereals, and Rice Products; 4 Fruits, Vegetables, and Produce; 5 Beverages; 6 Seafood; 8 Water and Water-Based Products; 11 Meat, Poultry, and Eggs; 24 Dairy and Milk Products; 27 Legumes and Pulses; 28 Nuts,

Seeds, and Their Butters; 29 Edible Fungi; 30 Culinary Oils and Fats; 31 Sweeteners and Syrups; 32 Condiments, Sauces, and Dressings; 33 Chocolate, Cocoa, and Confectionery; 34 Herbs, Spices, Seasonings, and Salt; 35 Prepared and Packaged Meals.

Ingestible and applied health products. Category 16 Dietary Supplements (Human); 25 Cannabis, Hemp, and CBD Products; 26 Peptides, SARMs, and Research Chemicals.

Personal care and cosmetics. Category 2 Infant and Child Personal Care; 13 Cosmetics and Personal Care, Leave-on; 14 Cosmetics and Personal Care, Rinse-off; 15 Feminine Care; 20 Oral Care.

Home and household. Category 12 Household Cleaning and Dishwashing; 19 Laundry and Fabric-Contact Home Products; 22 Home Air and Inhalation-Adjacent Products; 23 Food-Contact Consumer Goods and Kitchenware.

Children's durable and contact goods. Category 9 Infant and Child Contact Products; 10 Infant and Child Durable Goods and Textiles; 21 Children's Toys, Arts, and Crafts.

Pet products. Category 17 Pet Foods; 18 Pet Supplements.

Each category carries its own regulatory anchors, from FDA Closer to Zero and EU Regulation 2023/915 for infant foods, through CPSIA and ASTM F963 for children's durable goods, to the state cannabis testing programs where no uniform federal contaminant limit exists. A new product that does not fit an existing category routes through a defined novel-product protocol rather than being forced into an ill-fitting row.

3.3 Coverage in force

Category coverage is published subcategory by subcategory on the web as each subcategory's standard clears the publication gates, and a category is bound into a citable manual once every subcategory in it has a defensible standard. As of this edition the program has 348 subcategory standards in force across the 34 active categories, of which 320 already carry a firm per-analyte limit, together publishing 1,794 firm analyte limits. The remaining subcategory standards are in force as standards while one or more of their cells remain unpublished pending sufficient evidence, an explicit and disclosed state rather than a silent gap.

The certified analyte panel is ten analytes across the eight priority metals in every subcategory. Not every cell carries a firm number: where no government regulates a cell and the occurrence pool is not yet sufficient to publish, the cell is held rather than filled with a weak value, and the coverage figures above distinguish the standards in force from the cells carrying a firm limit for exactly this reason.

3.4 Manuals, web pages, and this program manual

The program publishes standards at three granularities that are the same content, not three sources of truth.

The **subcategory web pages** at heavymetalcertified.com are the finest granularity, one per subcategory, always reflecting the current edition, and best for lookup and search. The **category manuals** are the bound, citable volumes, one per category, each fixed at its revision date, carrying a DOI, and best for filings and exhibits; a brand is held to the standard in force when its product was tested, named on the certificate, not to a later revision. This **program manual** sits above both: it carries the framework the category manuals share and does not restate their per-subcategory numbers.

One category manual is currently in force, the Infant and Child Foods manual, 2026 Edition, published by the Institute of Contaminant Standards. Additional categories are published on the web ahead of their bound manuals, with manuals for Seafood, Grains, Dairy, Pet Foods, Culinary Oils, Butter and Ghee, and Herbs, Spices, and Salt in preparation. A category manual is superseded, never quietly corrected: a new edition receives a new DOI and the prior edition stays resolvable, so a filing that cited an earlier edition can always reach exactly what it cited, and any substantive change to a published limit runs through the Standards Ratchet of Part 2.11.

Part 4. Lot Testing and Surveillance

Certification is continuous, and the testing schedule is what makes it so. This part specifies how often a certified product is tested, how samples are taken and their integrity preserved, when a change forces an unscheduled test, and how a laboratory result becomes a status. It summarizes the HMTc Lot Testing Schedule, the controlled operating document that governs testing decisions; where the two differ, the Schedule governs. The schedule is built around three commitments: legal defensibility, so every test and decision rule withstands scrutiny; scientific rigor, so protocols reflect current analytical practice; and commercial proportionality, so testing intensity tracks risk rather than burdening compliant brands uniformly.

4.1 Product risk tiers

Testing frequency is a function of both a product's certification status and its risk tier. Risk tiers reflect how the product is consumed or used: how frequently, at what volume relative to body weight, and whether it can serve as a sole or dominant source of exposure for its population. A testing gap in a high-consumption, sole-source product creates far more exposure risk than the same gap in an occasional one, and the schedule tests accordingly.

Within the infant and child foods category, for example, infant formula and dry cereals are high-risk because they can be a sole or dominant food consumed in high volume relative to body weight; purées and mixed meals are moderate-risk, consumed regularly as part of a diversified diet; and juice and snacks are lower-risk, consumed occasionally. Each category carries its own risk-tier assignments on the same logic, and the

Program Operator may adjust a specific product's tier for factors such as production volume, historical results, or a known high-risk ingredient. A teething product made primarily from rice, a high-arsenic ingredient, may be elevated a tier for testing purposes. Adjustments are documented and communicated to the licensee.

4.2 Testing frequency by status and risk tier

The following table gives the minimum routine-surveillance frequency. The Program Operator may increase frequency at any time on a risk signal, an audit finding, or a trend. All testing uses the full panel with speciation where required.

Status	High-risk	Moderate-risk	Lower-risk	Hold-and-release	Independent sampling
A	Monthly	Quarterly	Quarterly	Not required	Yes, per schedule
B	Every lot or monthly, whichever is more frequent	Monthly	Monthly	Recommended	Yes, monthly
C (low, 150 to 175 percent)	Every lot	Every lot	Every lot	Mandatory	Yes, every lot
C (high, 175 to 200 percent)	Every lot	Every lot	Every lot	Mandatory	Yes, plus third-party sampling

Three terms carry the integrity of the schedule. **Independent sampling** means samples are collected and submitted by a party independent of the brand's production team, the Program Operator, a designated auditor, or another approved party, so that testing is demonstrably free of selection bias. **Hold-and-release** means a finished lot is physically quarantined and may not ship until results are received and the lot is confirmed to meet at least its applicable status criteria, with documented quarantine location, responsible personnel, and release authorization. **Third-party sampling** at high probation adds a fully independent second data point, directed by the Program Operator at the brand's expense, when a MaxPoL

between 175 and 200 percent calls for maximum assurance of data integrity.

Surveillance intensity therefore follows status: routine for A, elevated for B, and every-lot hold-and-release for C, with a Status C product on high probation also subject to program-directed third-party sampling. Trend monitoring runs underneath all of it: a Status A licensee performs formal trend analysis at least quarterly using control charts or equivalent methods, and a result reaching 80 percent of any limit triggers confirmatory testing and enhanced monitoring even though the result is technically compliant, with notification and speciation where a Tier 1 metal is involved.

4.3 Baseline testing

Program entry is the three-lot baseline of Part 2.3. Each of the three distinct-run lots is tested for the full panel, speciated where required, sampled as a composite across the lot, and documented under chain of custody. The three results are evaluated collectively and the initial status is assigned from the worst-performing lot. The expedited baseline via prior-data credit, also in Part 2.3, allows a brand with qualifying accredited prior data to satisfy up to two of the three lots and complete the baseline with one independently sampled new lot; for any analyte the completing lot establishes alone, the next two routine surveillance lots serve as confirmatory baseline points.

4.4 Reflex testing triggers

Reflex testing is any test outside the routine schedule, triggered by an event that could change a product's heavy-metal profile. A reflex result carries the same weight as a routine one and the same status consequences. The triggering events and their testing requirements are:

Trigger	Testing requirement
New ingredient, supplier, ingredient origin or grade	Full panel on the next 2 lots
New water source or treatment change	Full panel on the next 2 lots
New or repaired product-contact equipment	Full panel on the next lot

Trigger	Testing requirement
New packaging material or supplier, liner or closure change	Full panel on the next 2 lots, to assess migration over time
Formulation change, ingredient ratio change, addition or removal	Full panel on the next 2 lots
Process change to contact time, temperature, pH, or mechanical stress	Full panel on the next lot
Upstream supplier notifies of a sourcing or processing change	Full panel on the next lot using the affected ingredient
Any Tier 1 metal at or above 80 percent of limit on routine testing	Confirmatory test on the next lot, with speciation if applicable
Supplier-intelligence alert on a supplier or ingredient category	Full panel on the next lot using the flagged input

A reflex test uses the full panel, follows the standard sampling and chain-of-custody requirements, and if it reveals a nonconformance triggers the applicable status transition and corrective action immediately, wherever the product sits in the routine cycle.

4.5 Speciation

The form of a metal, not just its total concentration, determines its toxicological significance, and the schedule requires speciation where form matters.

Metal	Form of concern	Mandatory speciation	Reflex trigger
Arsenic	Inorganic arsenic (iAs)	All products containing rice or rice-derived ingredients	Total As reaches 50 percent of the total-metal limit
Mercury	Methylmercury (MeHg)	All products containing fish or marine ingredients	Total Hg reaches 50 percent of the total-metal limit
Chromium	Hexavalent chromium (Cr(VI))	Only where a credible pathway exists	Total Cr anomalously high, or Program Operator directs
Pb, Cd, Ni, Al, Sn	Total element only	Not required	Not applicable

When speciation is performed, the speciated result becomes the compliance result for that metal, superseding the total value. The reflex trigger of "approaches the limit" is operationally 50 percent of the applicable total-metal limit; the Program Operator may lower that threshold on a risk basis but may not raise it. Failure to perform required speciation renders the lot ineligible for certification.

4.6 Sampling and chain of custody

All certification testing is conducted on a lot basis, and each analytical sample is a composite prepared from multiple consumer-unit samples drawn from across the lot, designed to represent the lot's average quality with 95 percent confidence. Alternative sampling approaches require prior written approval.

A complete chain-of-custody record is required for every sample, documenting who collected it and their affiliation, when and where, how it was sealed, labeled, and transported, who received it at the laboratory, and confirmation of sample integrity on receipt. A documented breach, a custody gap, a broken or missing seal, non-conforming labeling, a transport non-conformance, or a receipt discrepancy, invalidates the result and is treated as a data-integrity issue, with the lot subject to retest under an approved confirmatory protocol and repeated breaches escalating to suspension. The Program Operator may exercise documented discretion on a clearly procedural lapse where substantive sample integrity is preserved, for example a timestamp recorded an hour late where the actual custody record is intact; the default disposition for any ambiguity is invalidation, and discretion is never exercised where sample integrity itself is in doubt.

Laboratory reports express results on the finished-product native basis, identify each analyte and its form, and include the limit of quantitation and the measurement uncertainty for each analyte, supplied as structured machine-readable data. A non-detect is treated as zero for compliance only where the limit of quantitation is at or below the program limit; where it is not, the result is indeterminate and a more sensitive method is required.

4.7 Decision rules and status mapping

Every lot result is evaluated under the measurement-uncertainty decision rule of Part 2.7: pass when the result plus uncertainty is at or below the limit, fail when the result minus uncertainty is above it, and borderline otherwise, with a borderline result resolved by confirmatory action before any status is assigned. Percent-of-limit is then computed for every analyte, and the maximum across analytes maps the lot to a status.

MaxPoL	Metal tier	Status	Action
0 to 100 percent	Any	A	Routine surveillance
101 to 150 percent	Tier 2 only	B	Corrective action in 10 days, elevated testing
101 to 150 percent	Any Tier 1 metal	C	Corrective action in 5 days, every-lot hold-and-release
151 to 175 percent	Any	C (low)	Corrective action in 5 days, every-lot hold-and-release
176 to 200 percent	Any	C (high)	Corrective action in 5 days, every lot plus third-party sampling
Above 200 percent	Any	D (suspension)	Stop-ship, corrective action in 2 days, mark removal

The prohibition on testing into compliance applies throughout: any retesting is part of a pre-approved, scientifically justified, documented confirmatory plan, and undocumented resampling or lab-shopping is a data-integrity violation subject to suspension or revocation.

Part 5. Methodology, Defensibility, and Governance

5.1 Why the methodology is defensible

A certification program that sets numeric limits is only as strong as its answer to three attacks: that the method is opaque and cannot be reproduced, that the policy choice behind it is arbitrary, and that its inputs were selected to reach a predetermined result. The program is built so that each attack fails on the merits.

The opacity attack fails because the reasoning is documented even though the engine code is not. The principle of standards-setting is in Part 3.1; the mechanism is in the Governing Principles; and the inputs, the source pages that feed each occurrence-set cell, are public on heavymetalindex.com. What is held as a trade secret is the pooling engine's implementation, not the reasoning, and holding it protects years of methodology development from a competing certifier without making the result unverifiable. The default method, moreover, is a government-maximum lookup, which is reproducible by anyone with the cited regulation in hand.

The arbitrariness attack fails because the policy choice traces to the five Governing Principles. An adversary who wants to argue that a different calibration would better serve those principles is welcome to make that argument on the merits, which is exactly the public discourse a credible standard invites. The program does not claim its calibration is the only defensible one; it claims the calibration best serves all five principles at once, and it publishes the pass rate the calibration produces rather than hiding it.

The selection-bias attack fails because the corpus is open. An adversary who believes a relevant source was omitted, or an irrelevant one included,

can check the routing audit and the source frontmatter and point to the specific page in dispute. Selection-bias arguments cannot be advanced in the abstract against an open corpus; they must name a page, and the naming is itself auditable.

Controlled-terms access. For the occurrence-set cells, the pooled distribution itself is available to qualified reviewers under controlled terms: regulatory bodies acting under official inquiry, litigation counsel where a certified product or the methodology is at issue, licensees auditing the methodology behind their own category, and academic researchers under an approved protocol. The Program Operator responds within thirty business days, providing the aggregate distribution directly to a regulator under official inquiry, or, to a non-regulatory requestor, under an agreement that preserves the engine's trade-secret status while making its inputs and outputs auditable. A refusal states its specific basis and is reviewable by the independent Appeals Committee. This design makes the evidence auditable while keeping the engine protected, and it is the structural answer to the "we set this limit because we said so" attack, which the program cannot be made to advance because its source attribution is public.

5.2 Program-level adversarial read

The following are the principal attacks a hostile reviewer brings against a program of this kind, and the program's answer to each. The list is not exhaustive; it is the set most likely to appear in a deposition, a regulatory comment, or a competitor's brief.

"The limits are self-published." They are not. The evidence base is an architecturally separate public corpus that the operator maintains but that does not endorse the program, and every occurrence-set limit traces to source pages a reader can audit. The default limit is a government maximum, not a program invention. Part 1.4 and Part 3.1 carry the detail.

"A four-metal panel is the industry norm; eight is marketing." The reverse. A four-metal panel is silent on four metals that reach food and consumer products through documented pathways. Publishing limits for all eight, and testing every certified product against ten analytes, closes a real gap rather than manufacturing one. Part 2.2 states the pathways.

“A limit sits above a legal ceiling.” It cannot. The health-based floor binds first at every cell: no published value exceeds the lowest applicable regulatory ceiling in the row’s basis, and the program certifies against the subcategory in native basis rather than against an element-wise envelope that could exceed a formulation-specific legal limit. Part 3.1 states the floor and Part 2.6 the native-basis rule.

“The pass rate proves the threshold was set to a commercial outcome.” Every percentile-based food-safety limit embeds a pass rate; the program’s distinction is that it publishes its own. The calibration operates only beneath the binding health-based ceiling, in the space where the literature alone does not fix a single number, and the pass rate is disclosed precisely so that it is discoverable rather than hidden. Part 3.1 makes the argument.

“The ratchet can be gamed.” The trigger combines Status A and Status B pools so best data cannot be withheld, caps any single licensee at 25 percent of the pool so no dominant brand can drive it, requires two consecutive annual reviews, caps the step at 20 percent, and closes the exit-and-re-enter pathway at twenty-four months. Each clause has a stated anti-gaming rationale. Part 2.11 specifies them.

“The operator judges its own paying customers.” The appeals process is two-tier, with a structurally independent Appeals Committee holding binding reversal authority, and conflict-of-interest and recusal rules govern who may decide. The independence of the second tier is the answer an internal-only process cannot give. Part 2.13 specifies the structure.

“The methodology is opaque.” The reasoning is documented and the inputs are public; only the engine implementation is a trade secret, and qualified reviewers may audit the engine’s inputs and outputs under controlled terms. Part 5.1 sets out the access path.

5.3 The recognition landscape

The program occupies a specific position among the certifications and standards a retailer or brand may already recognize, and it is useful to state the difference plainly rather than to claim the program is simply better.

Broad product certifications such as USDA Organic and the Non-GMO Project verify a production process or an input, not a finished-product contaminant level, and neither sets heavy-metal limits. This program is complementary to them: a product can be organic and still carry heavy metals, because the metals enter through soil, water, and equipment that an organic standard does not test for.

Contaminant-focused programs are the closer comparison. NSF/ANSI 173 certifies dietary supplements against contaminant limits; USP Verified covers supplement quality and identity; Informed Sport screens for banned substances with a heavy-metals component; the Clean Label Project tests and ranks products including for heavy metals; and ICH Q3D sets elemental-impurity limits for pharmaceuticals. This program differs from each on the same axes: it publishes the numeric limit before anything is certified against it, it certifies against the subcategory in native basis rather than a category-wide value, it tests a ten-analyte panel across every category rather than a narrower or category-specific one, it commits to a ratcheting limit rather than a static one, and it is backed by an architecturally separate, publicly auditable evidence base. The claim the program makes is narrow and checkable: not that a product is clean or safe, but that it was tested at an accredited laboratory at or below a published number on a stated panel in the form it is sold in.

5.4 Governance structure

The program is operated and governed by the Institute of Contaminant Standards. The Institute sets how limits are established and governs the program; it does not test products or sell marks itself, and that separation is what lets it arbitrate. Three bodies carry the governance.

The Institute of Contaminant Standards governs how limits are set and maintains both the certification program and the separate public evidence base. It does not itself publish threshold values or brand-level data into the public literature layer, which is the firewall that preserves the evidence base's independence.

The Technical Committee comprises toxicology, food-safety, and analytical-chemistry experts with documented independence from licensees. It reviews methodology, recommends reclassification of a metal

between tiers, and may recommend additions to the analyte panel. It advises; it does not render binding decisions.

The independent Appeals Committee holds binding review authority over contested certification decisions, as specified in Part 2.13. The Program Operator retains final authority on classification and methodology, subject to that review.

Substantive changes to a published limit run through the Standards Ratchet, which requires a stated trigger and caps how far a limit may move in one step. Limits tighten as evidence and analytical feasibility improve; they do not loosen because a brand found them inconvenient.

5.5 Glossary

Action level. A concentration limit that triggers sourcing, process, and quality-control intervention when exceeded. Not a safety threshold.

ALARA. As Low As Reasonably Achievable. The principle of continuous, feasible reduction of contaminant levels over time, treated as an enforceable obligation for metals with no known safe threshold.

Basis (native). The physical form a limit and a result are expressed in: powder as placed on market, ready-to-feed as consumed, dry as sold, purée as consumed. Sources in other bases are converted inward.

CAPA. Corrective and Preventive Action. The documented process of root-cause analysis, correction, prevention, and verification opened when a lot falls short.

Chain of custody. The unbroken documented record of a sample from collection through laboratory receipt and analysis. A documented breach invalidates the result.

Licensee. A brand or manufacturer under a license agreement authorizing participation and, on certification, use of the mark.

Lot. A quantity of product manufactured under uniform conditions, identified by common date, line, or shift codes.

Measurement uncertainty (MU). The laboratory's stated expanded uncertainty, typically at 95 percent confidence, associated with a reported result. Applied in the two-sided pass and fail decision rule.

Percent-of-limit (PoL) and MaxPoL. A result normalized to its limit, result divided by limit times 100. MaxPoL is the highest PoL across a lot's analytes and drives its status.

Rationale tag. The one label attached to every published limit: regulatory-alignment, literature-baseline, feasibility-driven, or precautionary.

Reflex testing. A test triggered outside the routine schedule by a change or a warning result, carrying the same weight and consequences as a routine test.

Speciation. Analysis that determines the specific chemical form of a metal, inorganic arsenic, methylmercury, or hexavalent chromium, which carries the toxicity. The speciated result supersedes the total when performed.

Tier 1 and Tier 2. The toxicological classification of the eight metals. Tier 1 (Pb, Cd, iAs, MeHg) admits no transitional exceedance; Tier 2 (Ni, Sn, Al, Cr) admits a transitional exceedance to 150 percent under conditions.

Trigger pool. The combined Status A and Status B lot results used to evaluate whether the Standards Ratchet fires for an analyte, capped so no single licensee exceeds a quarter of it.

Colophon

This manual is a compiled document. Its content is authored as source files, assembled into a single master for each edition by a build script, and typeset to PDF through a Typst book template. The internal and public editions are generated from one source: the public edition omits the internal Program Operations part and the internal-use notice, and the two never diverge on a fact. Regenerating the manual after a source change is a single build, which is what keeps the bound document and the operating reality from drifting apart.

The program's authoritative operating documents, the Program Governance Policy and the Lot Testing Schedule, are under separate version control and may be revised between editions of this manual. Where this manual and a controlled operating document disagree, the controlled document governs for operational purposes, and the Program Operator issues a Technical Clarification.

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References and Sources

The program's design rationale is set out in three peer-track papers, cited throughout this manual and available on Zenodo:

- Pendergrass, K. *The Counterproductive Consequences of Public Exposé Testing*. doi: 10.5281/zenodo.19470572.
- Pendergrass, K. *The Cost of Operating Without Credible Third-Party Heavy-Metal Certification*. doi: 10.5281/zenodo.18903738.
- Pendergrass, K. *Certification as a Framework for Reducing Heavy Metal Exposure in Infant and Child Foods*. doi: 10.5281/zenodo.18905821.

The occurrence and toxicology evidence underlying the program's limits is compiled, with per-source provenance, at the Heavy Metal Index (heavymetalindex.com). Each source page there resolves to an underlying study or agency document. The principal regulatory and toxicological instruments the program relies on, cited by their controlling identifiers on the relevant pages, include:

- United States Food and Drug Administration, Closer to Zero action plan, and its action levels for lead and inorganic arsenic in foods intended for babies and young children.
- European Commission Regulation (EU) 2023/915 on maximum levels for certain contaminants in food, and its amendments.
- Codex Alimentarius maximum levels for contaminants (CXS 193-1995) and its commodity standards.
- Joint FAO/WHO Expert Committee on Food Additives (JECFA) evaluations, including the withdrawal of the provisional tolerable weekly intake for lead and the tolerable intakes for aluminum and tin.
- European Food Safety Authority Scientific Opinions on lead (2010), cadmium, and the 2024 reassessment of inorganic arsenic in food, and the tolerable intakes for nickel and methylmercury.

- International Agency for Research on Cancer (IARC) Monographs classifying inorganic arsenic, cadmium and cadmium compounds, and hexavalent chromium as Group 1 carcinogens.
- United States House of Representatives, Subcommittee on Economic and Consumer Policy, 2021 report on toxic heavy metals in commercial baby foods.

Full bibliographic detail, sample sizes, methods, and basis for every occurrence source are on the corresponding source pages at heavymetalindex.com; a reader auditing a specific limit follows the category manual's sources-cited section to those pages. A structured reference export in an alternative bibliographic format is available from the Program Operator on request.